Laboratory 11

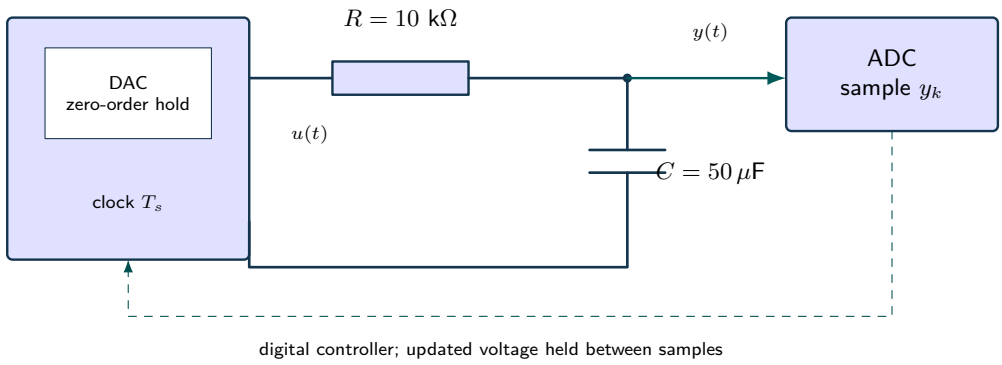
Sampling can destabilize a continuous-time design

Worked solutions – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Plant: $\dot{y} = -2y + 2u$. Nominal digital PI applies $u_k = 2(r_k - y_k) + I_k$ before updating $I_{k+1} = I_k + 4T_s(r_k - y_k)$. The exact hold and controller update order are part of the model.

1. A sampled controller connected to a continuous RC circuit

The physical example is an ideal 10 kΩ resistor and 50 μF capacitor, with held input voltage u and capacitor voltage y. An ideal ADC/DAC pair connects the circuit to a digital PI controller.



A held voltage drives an ideal RC network. The capacitor voltage is sampled by an ideal ADC; the software controller updates the DAC once per period. Quantization, converter lag and electrical loading are omitted.

(a) The plant is $\dot{y} = -2y + 2u$. Derive its exact update for input held constant over a sampling interval T_s, and evaluate coefficients at $T_s = 0.5$ s.

Worked reasoning. Integrating the scalar equation gives $y_{k+1} = e^{-2T_s}y_k + (1 - e^{-2T_s})u_k$. Thus $a = e^{-2T_s}$ and $b = 1 - a$. At 0.5 s, $a = e^{-1} \approx 0.367879$ and $b \approx 0.632121$. The update is exact at sample instants for held input and this continuous model; it does not approximate the exponential by forward Euler.

(b) Use $u_k = 2(r_k - y_k) + I_k$ and $I_{k+1} = I_k + 4T_s(r_k - y_k)$. Derive the state update matrix for $[y_k, I_k]^T$ and its reference-input vector.

Worked reasoning. Insert the current integral state into the plant update: $y_{k+1} = (a - 2b)y_k + bI_k + 2br_k$. The integral update is $I_{k+1} = -4T_s y_k + I_k + 4T_s r_k$. Hence $M = \begin{bmatrix} a - 2b & b \\ -4T_s & 1 \end{bmatrix}$ and $B_r = [2b, 4T_s]^T$. Computing control with I_{k+1} instead of I_k would produce a different matrix and potentially a different stability boundary.

(c) Derive the continuous-time PI characteristic polynomial for $k_p = 2$, $k_i = 4$ and compare its poles with what a sampled implementation must check separately.

Worked reasoning. The physical circuit has $RC = 0.5$ s and $G = 1/(0.5s + 1)$. With $C = 2 + 4/s$, the characteristic equation is $s(0.5s + 1) + 2s + 4 = 0$, or $0.5s^2 + 3s + 4 = 0.5(s + 2)(s + 4)$. Both continuous poles are stable. Sampling introduces the held-input update and a specific integral-update order; its eigenvalues must lie inside the unit circle. Continuous stability alone imposes no guarantee on an arbitrarily large sampling interval.

2. A stability proof instead of a sampling-rate guess

Derive the exact sampled characteristic polynomial and apply all second-order Jury conditions. Keep the strictness of the interval explicit.

(a) Derive the characteristic polynomial $z^2 - tz + d$ of the update matrix. Express t and d using $a = e^{-2T_s}$.

Worked reasoning. Since $b = 1 - a$, the upper-left entry is $3a - 2$. Thus $t = \operatorname{tr} M = 3a - 1$ and $d = \det M = 3a - 2 + 4T_s(1 - a)$. The polynomial is $z^2 - (3a - 1)z + [3a - 2 + 4T_s(1 - a)]$. Discrete stability requires both root magnitudes below one; negative real parts, the continuous-time criterion, are not sufficient here.

(b) Use the real second-order Jury conditions to prove strict stability for $0 < T_s < 0.75$ and failure for $T_s \geq 0.75$. Show the three inequalities rather than only checking selected samples.

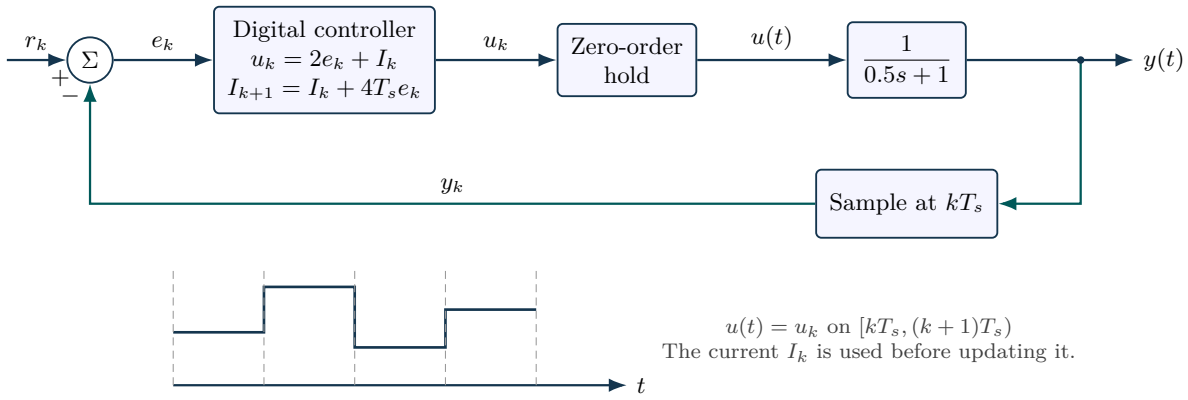
Worked reasoning. For $z^2 - tz + d$, the conditions are $1 - t + d > 0$, $1 + t + d > 0$ and $1 - d > 0$. They become $4T_s(1 - a) > 0$, $6a - 2 + 4T_s(1 - a) > 0$, and $(1 - a)(3 - 4T_s) > 0$. The third gives $T_s < 0.75$. For the middle condition, half its value is $f = (3 - 2T_s)e^{-2T_s} + 2T_s - 1$. For $0 < T_s \leq 0.5$, the bound $e^{-2T_s} > 1 - 2T_s$ gives $f > 2(1 - T_s)(1 - 2T_s) \geq 0$. For $0.5 \leq T_s < 0.75$, both terms in f are nonnegative and the exponential term is positive. Thus the exact strict interval is $(0, 0.75)$; the third inequality fails afterward.

(c) Retain $k_p = 2$ but allow positive k_i. Derive a necessary sampling bound from the determinant condition alone. Evaluate it at $k_i = 8$ and state why the other Jury inequalities are still needed.

Worked reasoning. Writing $a = e^{-2T_s}$, the determinant becomes $d = 3a - 2 + k_i T_s(1 - a)$. Therefore $1 - d = (1 - a)(3 - k_i T_s) > 0$ requires $T_s < 3/k_i$ for positive T_s. At $k_i = 8$, this gives $T_s < 0.375$ s. It is only one of three strict Jury conditions; the remaining conditions involving $1 - t + d$ and $1 + t + d$ must also hold. A necessary bound is not automatically a complete stability interval for every gain.

3. The onset of sustained sampled oscillations

Compare stable, marginal and unstable sample periods. Interpret the boundary in physical time, not only as a point in the discrete z-plane.



The sampler, digital update order and hold are explicit. Continuous plant dynamics and sampled controller states occupy different parts of the loop.

(a) At $T_s = 0.75$, find determinant, trace and the nature of the poles. Explain why this endpoint is excluded even though no pole magnitude exceeds one.

Worked reasoning. Here $a = e^{-1.5}$ and $d = 3a-2+3(1-a) = 1$. The trace is $t = 3e^{-1.5}-1 \approx -0.330610$. Since $|t| < 2$, the discriminant $t^2 - 4$ is negative. The conjugate roots have product one, hence each magnitude is one. Generic initial conditions retain an oscillation instead of decaying. Strict asymptotic stability therefore excludes the endpoint.

(b) Compute the spectral radius for $T_s = 0.05$, 0.5 and 1 s. Classify each implementation and relate the result to the analytic interval.

Worked reasoning. For each interval form M from Problem 1(b) and take $\rho(M) = \max|\lambda_i(M)|$. The values are approximately 0.893907, 0.606531 and 1.365534. The first two are below one and lie inside the analytic stability interval. The 1 s implementation is unstable. A smaller per-sample radius alone does not compare settling time in seconds because different samples represent different physical durations.

(c) At $T_s = 0.75$ s, derive the oscillation angle per sample and corresponding angular frequency per second from the conjugate unit-circle pair.

Worked reasoning. At the boundary, $d = 1$ and $t = 3e^{-1.5} - 1$. Write the roots as $e^{\pm j\psi}$, so $2\cos\psi = t$ and $\psi = \cos^{-1}[(3e^{-1.5} - 1)/2]$. The sampled oscillation frequency is $\omega_d = \psi/0.75$ rad/s. The amplitude does not decay for generic excitation of these modes. This frequency belongs to the implemented discrete loop, not to a marginal continuous PI design.

4. An equilibrium is not a convergence guarantee

The sampled controller has a nominal reference equilibrium even at sample periods that make it unstable. Separate equilibrium equations from local behavior near that equilibrium.

(a) For a mode with discrete magnitude ρ, define an equivalent envelope decay rate per second. Compare the two stable sampling periods in Problem 3(b) and explain why this comparison still does not fully describe their transients.

Worked reasoning. After k samples the envelope is proportional to $\rho^k = e^{(\ln \rho)t/T_s}$. The decay rate is $\gamma = -\ln \rho / T_s$. For 0.05 s it is about 2.243 s^{-1}; for 0.5 s it is exactly 1 s^{-1} because $\rho = e^{-1/2}$. Despite the smaller per-sample radius at 0.5 s, its slow envelope decays more slowly in physical time. Pole angles, zeros, initial conditions and intersample behavior also influence the response.

(b) For constant reference r and any positive sampling interval, derive the nominal equilibrium of the discrete update. Explain why existence of this equilibrium does not ensure convergence to it.

Worked reasoning. The integral equation at equilibrium gives $4T_s(r - y) = 0$, hence $y = r$. The plant equilibrium satisfies $y = ay + bu$ and $b = 1 - a > 0$, so $u = y = r$. Since proportional error is zero, $I = u = r$. These algebraic values exist even when M is unstable. Convergence depends on its eigenvalues; a generic perturbation grows for $T_s > 0.75$ despite the formally correct equilibrium.

(c) At $T_s = 1$ s and constant r, initialize exactly at $[y, I] = [r, r]$. Explain why a perfect simulation can remain there forever although the implementation is unstable, and design a revealing numerical test.

Worked reasoning. The equilibrium satisfies the update algebraically, so exact arithmetic leaves it unchanged. Stability instead asks whether nearby trajectories return. At $T_s = 1$, the spectral radius is about 1.365534, so generic arbitrarily small perturbations grow. Run both the exact-equilibrium case and cases with small state perturbations, report the state-error norm and eigenvalues, and check that roundoff is not being mistaken for a controlled disturbance. An uneventful equilibrium trajectory cannot establish stability.

5. Implementation order and what happens between samples

Two programs can implement the same displayed PI gains but update the integrator in different orders. Compare the resulting matrices and reconstruct the actual continuous circuit motion.

(a) Now update $I_{k+1} = I_k + 4T_s(r_k - y_k)$ first and apply $u_k = 2(r_k - y_k) + I_{k+1}$ during the next plant interval. Derive the new matrix and show directly that $T_s = 0.5$ is unstable.

Worked reasoning. Substitution yields $M_{new} = \begin{bmatrix} a - b(2 + 4T_s) & b \\ -4T_s & 1 \end{bmatrix}$. At $T_s = 0.5$, $a = e^{-1}$ and $b = 1 - a$. Its trace is $5a - 3$ and determinant is $3a - 2$. The Jury quantity $1 + t + d = 8a - 4$ is negative because $a < 1/2$. Hence this implementation is unstable at 0.5 s, although the original update order was stable. Identical gain labels do not specify a digital controller fully.

(b) Given y_k and held input u_k, derive $y(kT_s + \tau)$ for $0 \le \tau \le T_s$. Explain why a plot of sample values alone may conceal an implementation error or performance issue.

Worked reasoning. The continuous solution within the interval is $y(kT_s + \tau) = e^{-2\tau} y_k + (1 - e^{-2\tau}) u_k$. At $\tau = T_s$ it must match the exact discrete update. This identity checks held-input simulation and indexing. Sample-only plots omit what happens between samples and may hide an incorrectly applied hold or control-update time. For this scalar first-order plant with constant input, each interval is monotone between its endpoint values; more general plants can have intersample peaks.

(c) For this first-order circuit under a truly held input, prove whether an intersample extremum can exceed both endpoint values. Explain why that result does not generalize automatically to a higher-order plant.

Worked reasoning. Within one hold, $y(kT_s + \tau) = u_k + (y_k - u_k)e^{-2\tau}$. Its derivative is $-2(y_k - u_k)e^{-2\tau}$, whose sign is constant unless it vanishes identically. Thus the ideal first-order trajectory is monotone between samples and cannot have an interior peak beyond both endpoints. Higher-order dynamics, non-held actuation or hidden states can invalidate this property. Sampling-only plots can still hide the wrong update order or an incorrect hold implementation.